Beyond the Wronskian

Necessary vs. Sufficient Conditions for Linear Independence of ODE Solutions

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Outline

Motivation

What Got Me Thinking

Standard result (Kreyszig §2.8; Ross §4.1):

Sufficient condition

If $W[f_1, f_2](x_0) \neq 0$ for some $x_0 \in I$, then f_1, f_2 are **linearly independent** on I.

Kreyszig §2.8, Example 2

The condition $W(u_1, u_2) = 0$ is only *necessary* for linear dependence, not sufficient.

Natural question:

If $W(x) = 0$ *everywhere* on I, must f_1, f_2 be linearly *dependent*?

Surprising answer: No!

This project traces exactly why — and why it does not break ODE theory.

Mathematical Preliminaries

Linear Dependence and Independence

Definition — Linear Dependence

Functions $f_1, \ldots, f_n$ on I are **linearly dependent** on I if $\exists\, c_1, \ldots, c_n$ *not all zero* such that

$$c_1 f_1(x) + \cdots + c_n f_n(x) = 0 \quad \forall\, x \in I.$$

Otherwise they are **linearly independent**.

Key point for two functions

f_1 and f_2 are linearly dependent **iff** one is a *constant multiple* of the other **on the entire interval** — not just at a single point.

The Wronskian

Definition (two functions)

For differentiable f_1, f_2 on I:

$$W[f_1, f_2](x) = \begin{vmatrix} f_1(x) & f_2(x) \\ f_1'(x) & f_2'(x) \end{vmatrix} = f_1 f_2' - f_1' f_2.$$

General definition (n functions)

$$W[f_1, \ldots, f_n](x) = \begin{vmatrix} f_1 & \cdots & f_n \\ f_1' & \cdots & f_n' \\ \vdots & \ddots & \vdots \\ f_1^{(n-1)} & \cdots & f_n^{(n-1)} \end{vmatrix}$$

The Wronskian is a **function of** x. Whether it is zero or nonzero — at a point, or everywhere on I — is the central question.

The Sufficient Condition

$W \neq 0 \Rightarrow$ Independence

Theorem (Ross Thm 4.4; Kreyszig §2.8)

If f_1, f_2 are differentiable on I and $W[f_1, f_2](x_0) \neq 0$ for some $x_0 \in I$, then f_1, f_2 are ***linearly independent*** *on I.*

Proof by contrapositive. Suppose f_1, f_2 are linearly *dependent*: $\exists\, c_1, c_2$ (not both 0) with

$$c_1 f_1(x) + c_2 f_2(x) = 0, \qquad c_1 f_1'(x) + c_2 f_2'(x) = 0 \quad \forall x \in I.$$

Since a non-trivial solution (c_1, c_2) exists, the coefficient determinant must be zero:

$$W[f_1, f_2](x) = f_1 f_2' - f_1' f_2 = 0 \quad \forall x \in I. \qquad \blacksquare$$

Quick Example: $\sin x$ and $\cos x$

$$W[\sin x,\, \cos x](x) = \sin x \cdot (-\sin x) - \cos x \cdot \cos x = -\sin^2 x - \cos^2 x = -1.$$

Since $W = -1 \neq 0$ everywhere, $\sin x$ and $\cos x$ are **linearly independent on every interval**. ✓

The natural next question

We have shown: dependent $\Rightarrow W \equiv 0$.

Is the converse true? $W \equiv 0 \overset{?}{\Rightarrow}$ dependent

The Counterexample

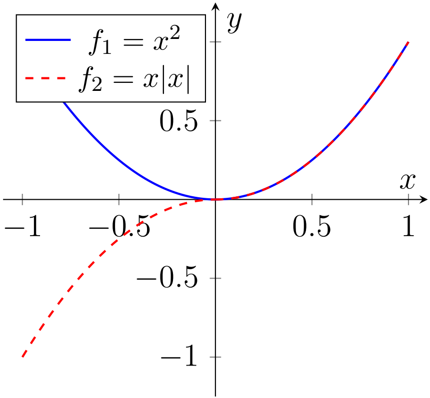
A Pair of Functions with $W \equiv 0$ but Linearly Independent

On $I = [-1, 1]$, define:

$$f_1(x) = x^2, \qquad f_2(x) = x\,|x| = \begin{cases} x^2, & x \geq 0, \\ -x^2, & x < 0. \end{cases}$$

Observations:

- Both differentiable on I; $f_2'(x) = 2|x|$ (continuous).
- On $x \geq 0$: $f_1 = f_2 = x^2$.
- On $x < 0$: $f_1 = x^2$, $f_2 = -x^2$.
- *No single constant multiple works on all of I.*



Proving Linear Independence

Suppose $c_1 x^2 + c_2 x\,|x| = 0$ for all $x \in [-1, 1]$.

For $x > 0$: $f_2 = x^2$

$$(c_1 + c_2)\,x^2 = 0 \;\Rightarrow\; c_1 + c_2 = 0.$$

For $x < 0$: $f_2 = -x^2$

$$(c_1 - c_2)\,x^2 = 0 \;\Rightarrow\; c_1 - c_2 = 0.$$

Adding and subtracting: $2c_1 = 0$ and $2c_2 = 0$.

$$\boxed{c_1 = c_2 = 0.}$$

$\therefore$ the only solution is the trivial one $\Rightarrow$ f_1 **and** f_2 **are linearly independent on** $[-1, 1]$. $\blacksquare$

The Wronskian Vanishes Identically

Recall $f_1'(x) = 2x$ throughout. Compute W piecewise:

Case $x \geq 0$: $f_2 = x^2,\ f_2' = 2x$

$$W = x^2 \cdot 2x - 2x \cdot x^2 = 2x^3 - 2x^3 = 0.$$

Case $x < 0$: $f_2 = -x^2,\ f_2' = -2x$

$$W = x^2(-2x) - 2x(-x^2) = -2x^3 + 2x^3 = 0.$$

Both branches give $W = 0$, and $W(0) = 0$ trivially.

$$\boxed{W[f_1,\ f_2](x) \equiv 0 \text{ on } [-1, 1].}$$

We have **linearly independent** functions with a **vanishing Wronskian**. The converse fails.

What This Tells Us — Summary So Far

Statement	**True for arbitrary differentiable functions?**
$W \neq 0$ at some point $\Rightarrow$ independent	**YES** (Theorem 3.1)
Dependent $\Rightarrow W \equiv 0$	**YES** (proof in §3)
$W \equiv 0 \Rightarrow$ dependent	**NO** — counterexample above

Conclusion

$W = 0$ is a **necessary** consequence of dependence, but is **not sufficient** to guarantee dependence.

Yet in every ODE problem in class, $W \equiv 0$ does mean dependence. Why?

Abel's Identity and the Resolution

The Resolution: ODE Solutions Are Special

The counterexample functions x^2 and $x|x|$ *do* satisfy an ODE:

$$y'' - \frac{3}{x}\,y' + \frac{3}{x^2}\,y = 0.$$

But the coefficient $p(x) = -3/x$ **blows up at** $x = 0$, violating the *continuity* hypothesis required for Abel's Identity.

The critical distinction

These functions satisfy *some* ODE — just not one with continuous coefficients. When the coefficients *are* continuous, the loophole closes completely.

Abel's Identity

Theorem (Kreyszig §2.8; Ross Thm 4.5)

Let y_1, y_2 *be any two solutions of*

$$y'' + p(x)\,y' + q(x)\,y = 0,$$

where p, q *are* ***continuous*** *on an open interval* I*. Then for any* $x_0 \in I$*:*

$$\boxed{W(x) = W(x_0)\,\exp\!\left(-\int_{x_0}^{x} p(t)\,\mathrm{d}t\right).}$$

Since the exponential factor is **never zero**, the sign of $W(x)$ is determined entirely by $W(x_0)$ — this gives the all-or-nothing property.

Abel's Identity — Consequences

If ...	**Then W is ...**	**Consequence**
$W(x_0) \neq 0$ for some x_0	never zero on I	solutions **independent**
$W(x_0) = 0$ for some x_0	identically 0 on I	solutions **dependent**

Upgrade

For ODE solutions, the Wronskian test goes from merely *sufficient* to **necessary-and-sufficient**.

Proof Sketch of Abel's Identity

Differentiate $W = y_1 y_2' - y_1' y_2$:

$$W' = y_1 y_2'' - y_1'' y_2 \quad \text{(the } y_1' y_2' \text{ terms cancel).}$$

Since each y_i solves the ODE, substitute $y_i'' = -p\, y_i' - q\, y_i$:

$$W' = y_1(-p\, y_2' - q\, y_2) - (-p\, y_1' - q\, y_1) y_2 = -p(y_1 y_2' - y_1' y_2) = -p\, W.$$

So W satisfies the separable ODE

$$W' = -p(x)\, W \quad \Longrightarrow \quad W(x) = W(x_0)\, e^{-\int_{x_0}^{x} p(t)\, \mathrm{d}t}. \qquad \blacksquare$$

This is Abel's Identity. The exponential is never zero, so W is either always zero or never zero — the ***all-or-nothing property****.*

The Equivalence Theorem

Abel's Identity upgrades the Wronskian test for ODE solutions from *sufficient* to **necessary-and-sufficient**:

Theorem (Kreyszig Thm 2, §2.8; Ross Thms 4.4–4.5)

Under the continuity hypothesis on p and q:

$$y_1, y_2 \text{ linearly dependent } \iff W[y_1, y_2] = 0 \text{ everywhere on } I.$$

$$y_1, y_2 \text{ linearly independent } \iff W[y_1, y_2] \neq 0 \text{ everywhere on } I.$$

Key contrast with Section 4

Arbitrary differentiable functions are not constrained by any ODE, so Abel's Identity need not hold for them — and indeed it does not, as our counterexample shows.

Illustration: $\sin x$ and $\cos x$ for $y'' + y = 0$

Here $p(x) = 0$, so Abel's Identity gives:

$$W(x) = W(x_0) \cdot e^0 = \text{const.}$$

Indeed:

$$W[\sin x, \cos x] = -1.$$

- $W = -1 \neq 0$ everywhere $\Rightarrow$ **linearly independent** on every interval.
- The Wronskian is *constant* — does not vary with x.

Solutions of $y'' + y = 0$

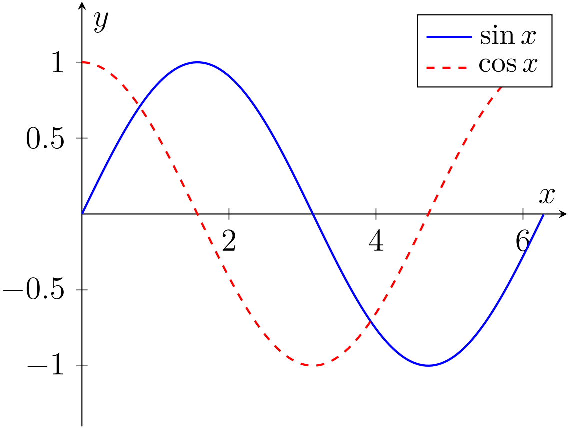


Illustration: The All-or-Nothing Property

Key takeaway from $\sin x$ and $\cos x$

This is the cleanest illustration of the **all-or-nothing property**: the Wronskian does not even change with x, let alone vanish.

Contrast with the counterexample

	x^2 and $x\|x\|$	$\sin x$ and $\cos x$
ODE coefficients	discontinuous at 0	continuous everywhere
Wronskian	$\equiv 0$	$\equiv -1$
Independent?	**Yes**	**Yes**
Abel applies?	**No**	**Yes**

Conclusion

Complete Picture

Condition	**Arbitrary functions**	**ODE solutions** (p, q continuous)
$W \neq 0 \Rightarrow$ independent	✓	✓
Dependent $\Rightarrow W \equiv 0$	✓	✓
$W \equiv 0 \Rightarrow$ dependent	× *(counterexample)*	✓ *(Abel)*

Three Takeaways

1. **The counterexample is instructive, not pathological.**
   x^2 and $x|x|$ are perfectly natural functions. What makes them special is that they are *not* solutions of any ODE with continuous coefficients — and that is exactly what the counterexample is illustrating.

2. **Abel's Identity is the key theorem.**
   It forces the Wronskian of ODE solutions to be either always zero or never zero — giving us a necessary-and-sufficient test.

3. **Hypotheses matter.**
   The continuity of $p(x)$ and $q(x)$ is not cosmetic. Removing it invalidates the conclusion, as the counterexample shows.

References

1. Kreyszig, E. (2011). *Advanced Engineering Mathematics* (10th ed.). Wiley.
   Sec. 2.8: Existence and Uniqueness, pp. 87–93.
2. Ross, S. L. (1984). *Differential Equations* (3rd ed.). Wiley.
   Sec. 4.1: Basic Theory of Linear Differential Equations, pp. 102–124.

Thank you!

Questions welcome.